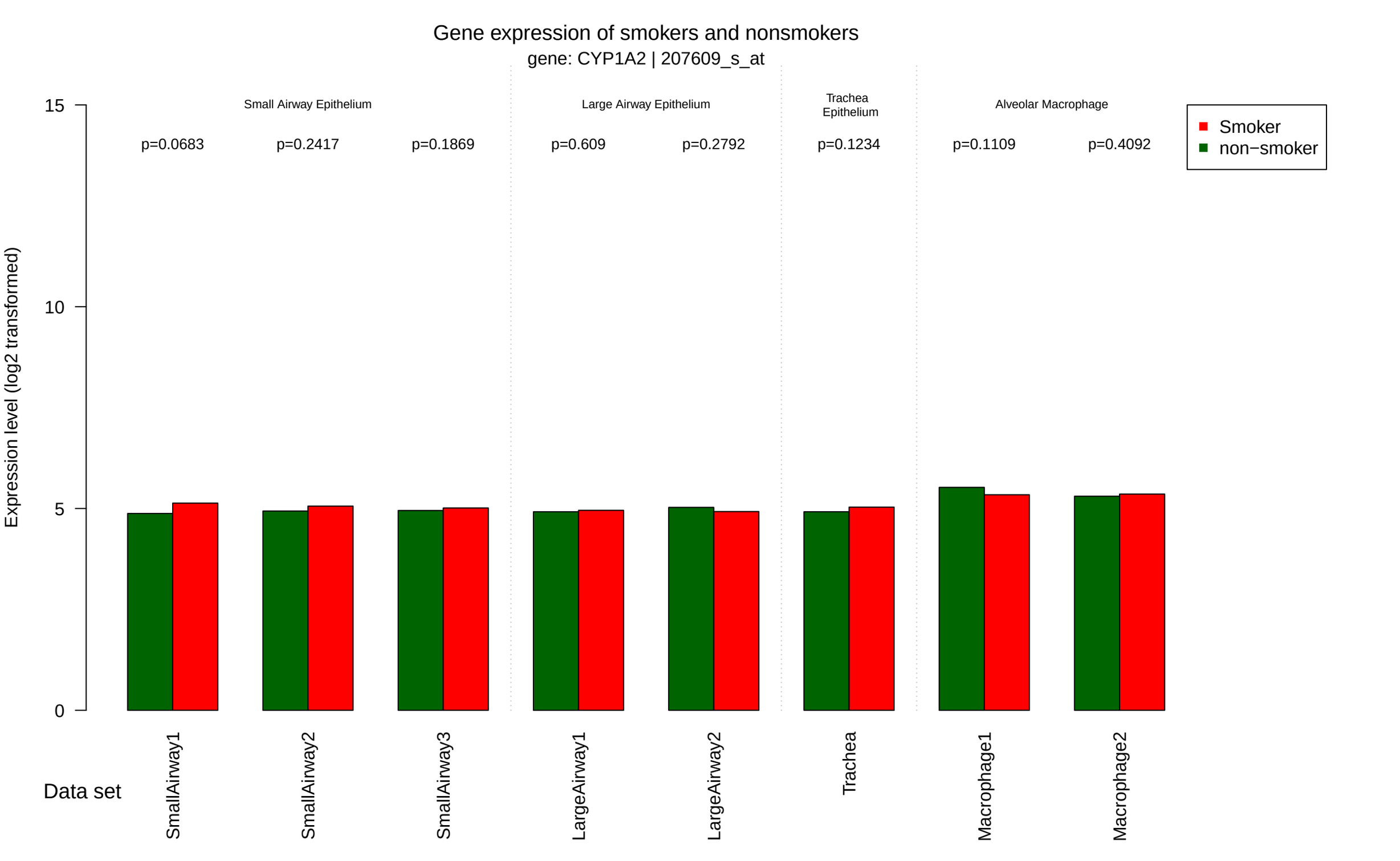
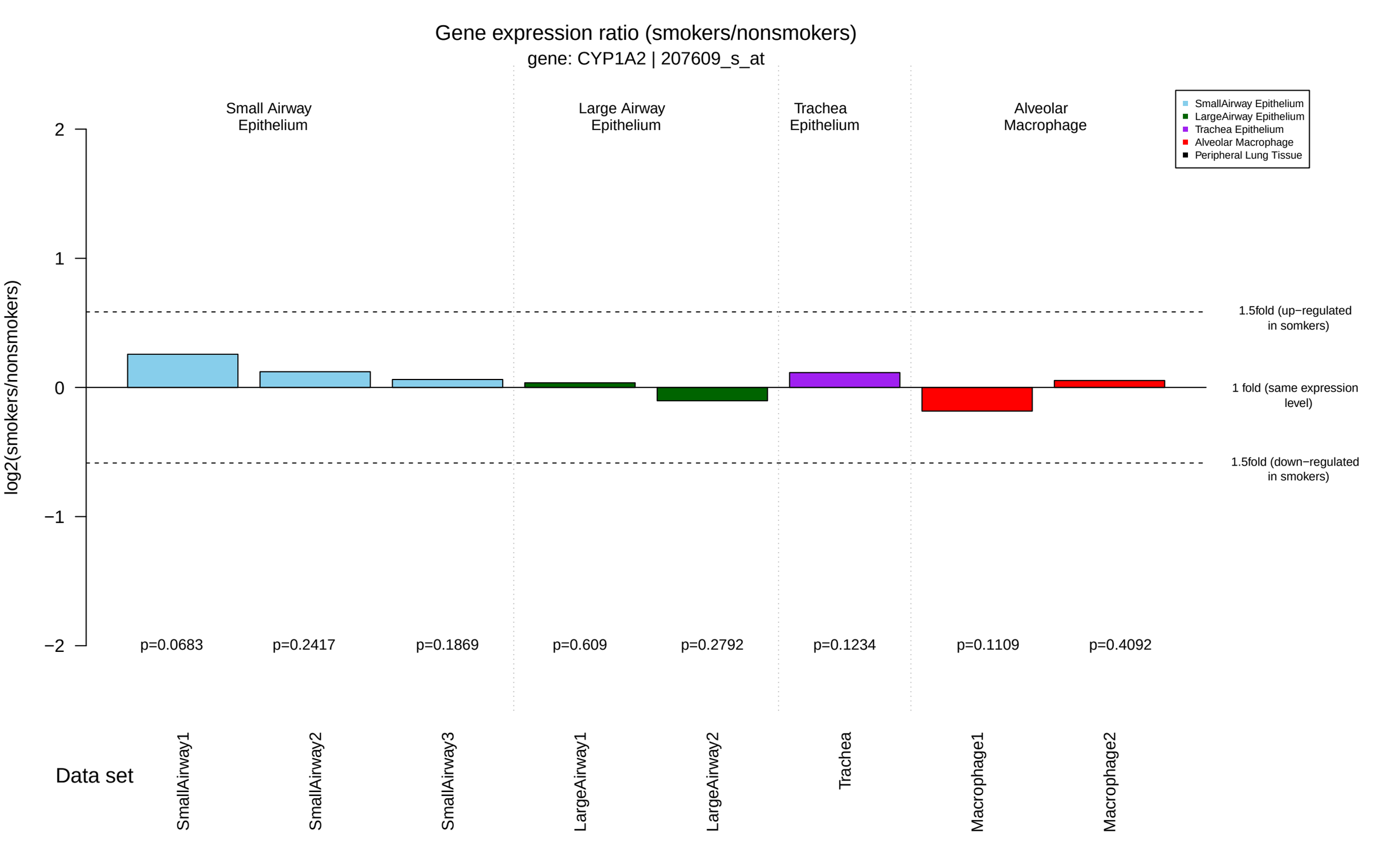
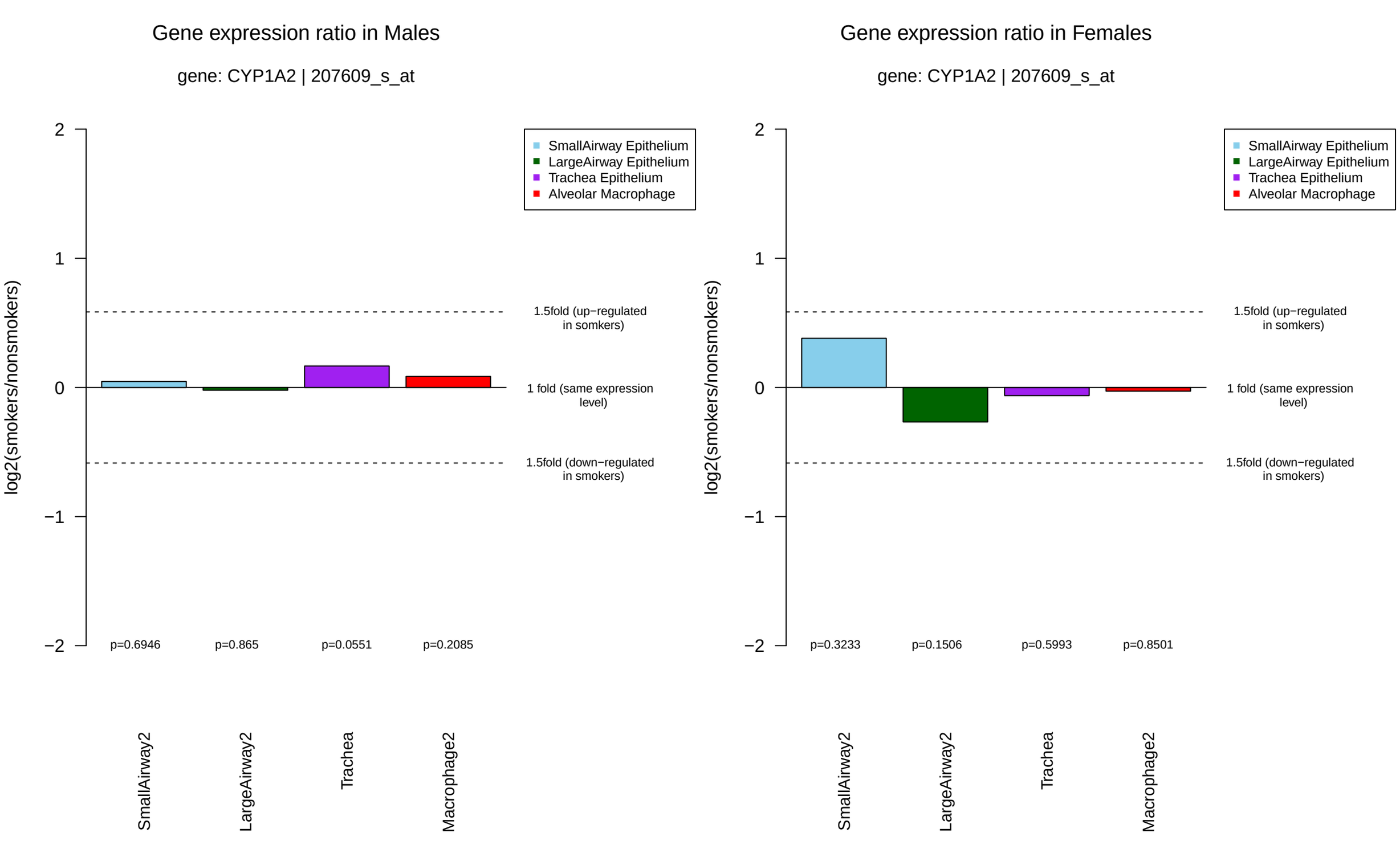
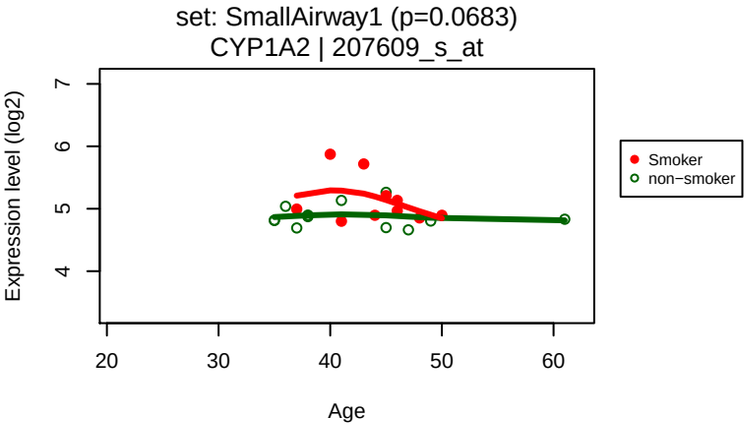
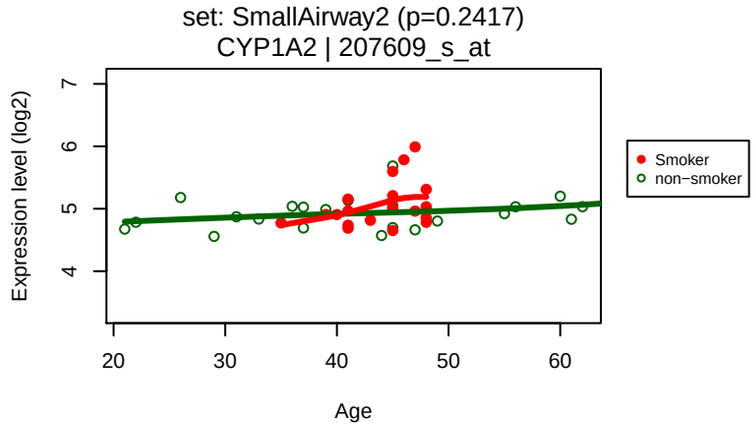
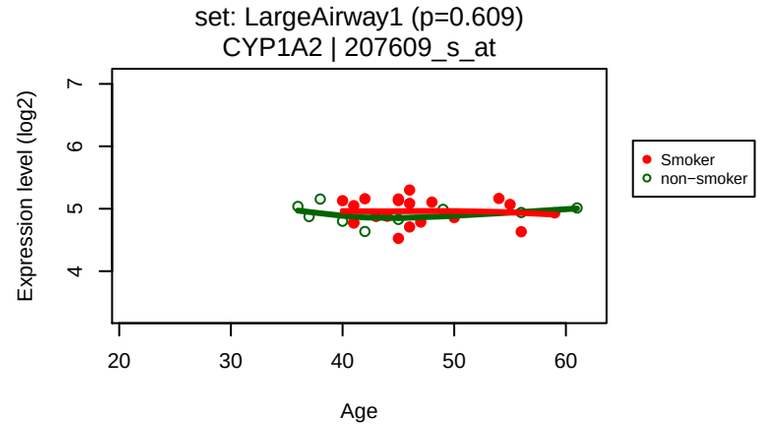

Gene expression of smokers and nonsmokers
gene: CYP1A2 | 207609_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0683
p=0.2417
p=0.1869
p=0.609
p=0.2792
p=0.1234
p=0.1109
p=0.4092
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

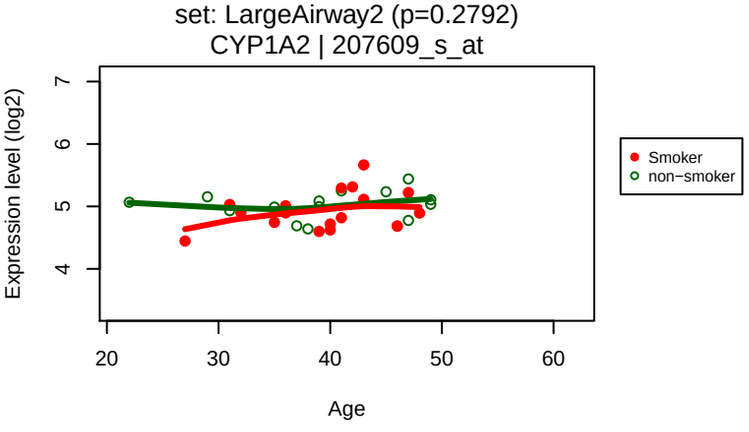

Gene expression ratio (smokers/nonsmokers)
gene: CYP1A2 | 207609_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0683
p=0.2417
p=0.1869
p=0.609
p=0.2792
p=0.1234
p=0.1109
p=0.4092
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

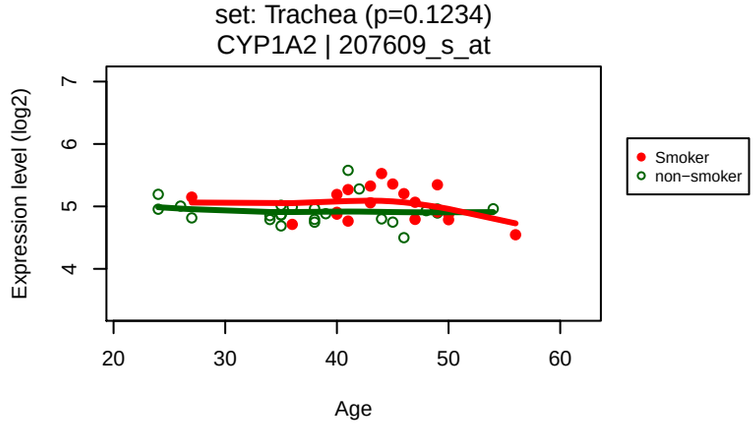

Gene expression ratio in Males
gene: CYP1A2 | 207609_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6946
p=0.865
p=0.0551
p=0.2085
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP1A2 | 207609_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3233
p=0.1506
p=0.5993
p=0.8501
SmallAirway2
LargeAirway2
Trachea
Macrophage2

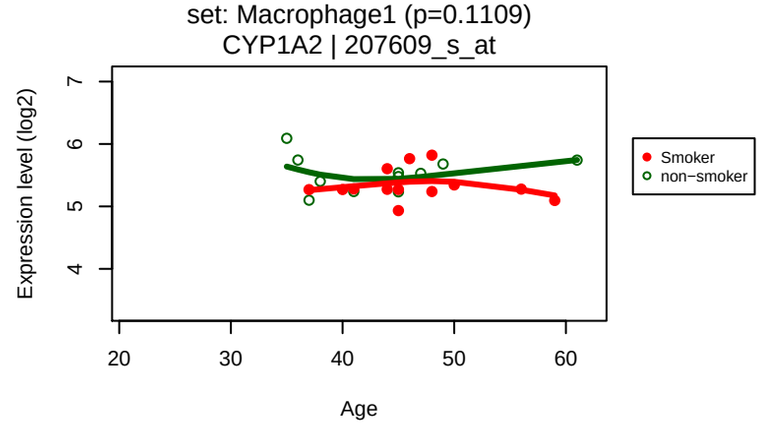

set: SmallAirway1 (p=0.0683)
CYP1A2 | 207609_s_at
Expression level (log2)
Age
Smoker
non−smoker

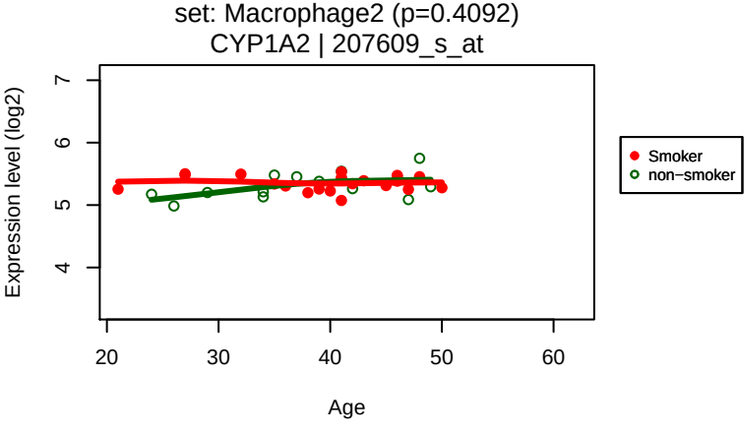

set: SmallAirway2 (p=0.2417)
CYP1A2 | 207609_s_at
Expression level (log2)
Age
Smoker
non−smoker

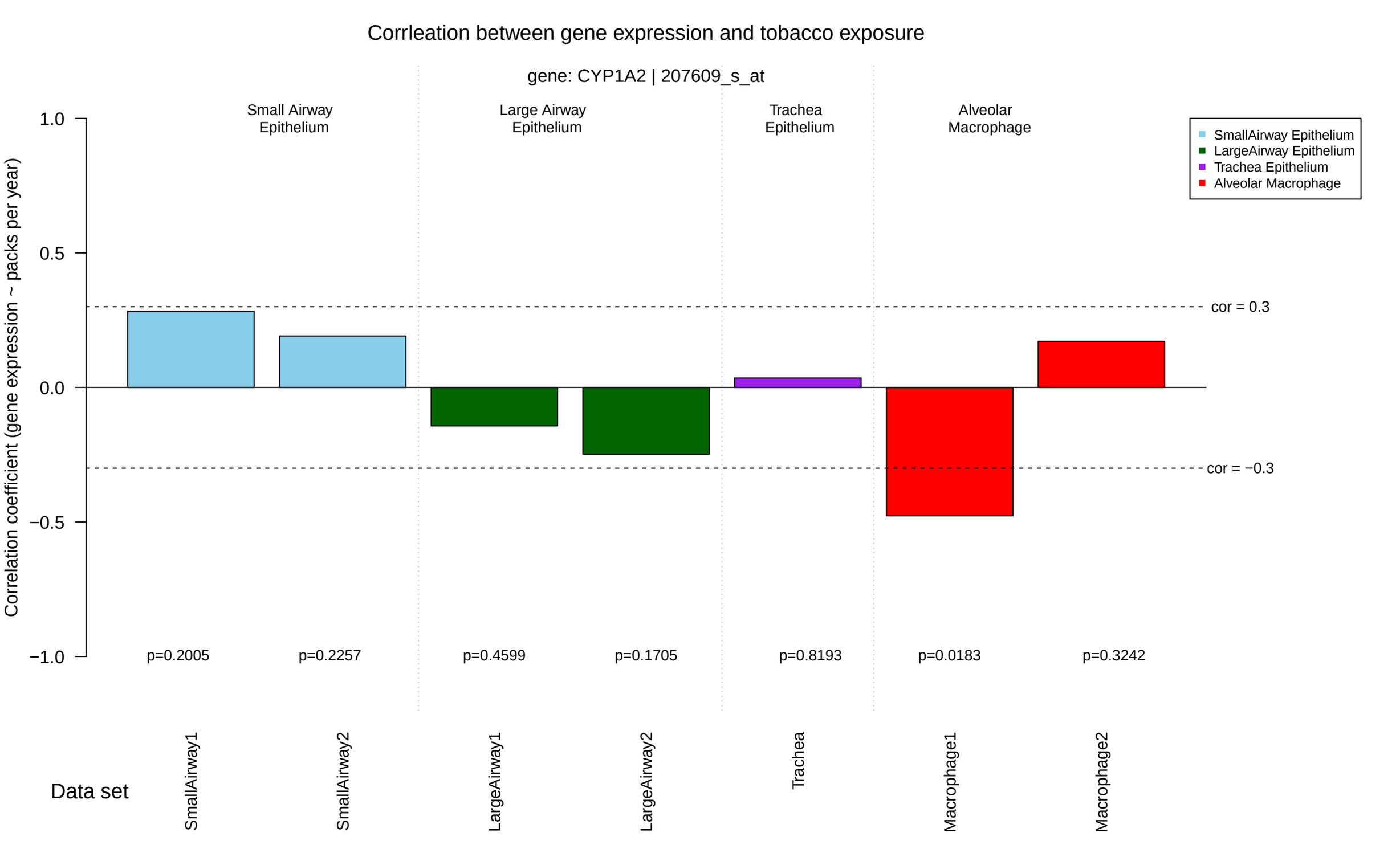

set: LargeAirway1 (p=0.609)
CYP1A2 | 207609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2792)
CYP1A2 | 207609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1234)
CYP1A2 | 207609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1109)
CYP1A2 | 207609_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4092)
CYP1A2 | 207609_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP1A2 | 207609_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2005
p=0.2257
p=0.4599
p=0.1705
p=0.8193
p=0.0183
p=0.3242
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2